

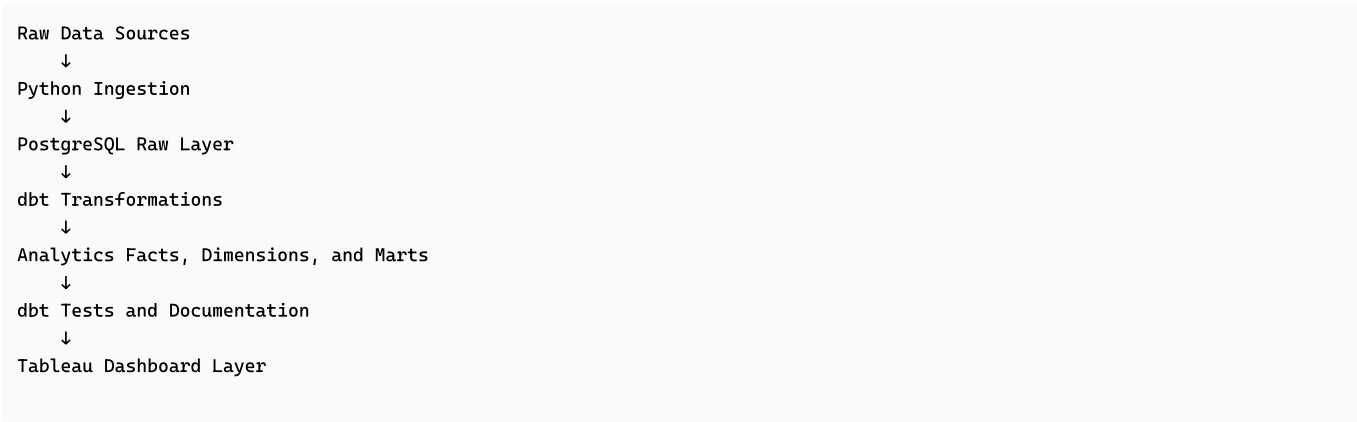
Search & Ads Growth Analytics Platform Architecture Document

1. Project Overview

The Search & Ads Growth Analytics Platform is an end-to-end analytics engineering project designed to simulate how product analytics systems operate in a real business environment.

The project focuses on clickstream behaviour, funnel performance, revenue attribution, user-level engagement, and simulated A/B experimentation reporting.

Instead of building only a dashboard, this project builds the full data flow behind the dashboard:



The objective is to show how raw behavioural data can be transformed into reliable, documented, and dashboard-ready analytics tables.

2. Business Problem

A digital commerce or search platform needs to understand:

- How users move through the product funnel
- Where users drop off between view, cart, and purchase
- Which products, categories, and brands generate revenue
- How user engagement differs across sessions
- Whether a simulated product experiment shows positive or negative lift
- How daily growth KPIs change over time

The platform answers these questions by creating a structured analytics warehouse with tested dbt models and dashboard-ready marts.

3. Data Sources

- User event data containing views, carts, purchases, products, prices, users, and sessions.
- Product metadata source containing product, category, and brand information.

4. Tech Stack

- Raw clickstream event source
- Product master source system
- Central analytics warehouse
- Ingestion and source synchronization
- CSV processing and validation
- Database connectivity
- Transformations, tests, and documentation
- Version control
- dbt parse validation
- Dashboard and reporting layer
- Project documentation

5. Architecture

